Abhishek Ravikumar

+1 602-349-2403 | abhishek270102@gmail.com | linkedin.com/in/abhishek-ravikumar27 | github.com/abhir271 | Tempe, Arizona

Summary

Data professional with a Master's in Data Science and Analytics Engineering, focused on building reliable data pipelines, analytics and LLM powered systems. Hands on with Python, SQL, PostgreSQL and Spark, and with turning raw data into dashboards (Tableau, Power BI), APIs and cloud deployed services using AWS and CI/CD. Seeking Data Analyst and Data Engineer roles.

Education

Arizona State University — Tempe, Arizona
Master of Science in Data Science and Analytics Engineering; GPA: 4.0/4.0 — *May 2026*
- **Coursework:** Machine Learning, Statistical Modeling, Data Mining, Data Engineering, Distributed Computing, Deep Learning

Reva University — Bengaluru, Karnataka, India
Bachelor of Technology in Computer Science and Engineering; GPA: 3.6/4.0 — *June 2024*

Experience

Graduate Research Assistant — August 2026 – Present
Arizona State University × Honeywell — *Tempe, Arizona*
- Integrated **OpenAI** and **Anthropic Claude** APIs into a **LangChain**-orchestrated Python pipeline to parse aviation safety notices, achieving **99% recall** across **10,000+** records.
- Tuned **PostgreSQL** queries and indexing, cutting p95 latency by **20%**.
- Automated CI/CD via **GitHub Actions** on **AWS**, enabling **100+** peer code reviews with zero regressions across **6+** sprint releases.

Data Analyst — June 2025 – May 2026
Datasoothra — *Tempe, Arizona*
- Performed exploratory data analysis on large structured datasets with **Python (Pandas, NumPy)** to uncover trends and business insights.
- Designed and automated reporting dashboards with **Excel** and **Python**, improving reporting efficiency by **30%**.
- Wrote **SQL** queries (JOINs, subqueries, aggregations) to extract, transform and analyze relational data for business reporting.

Data Analyst Intern — October 2022 – May 2024
MEVI Technologies LLP — *Bengaluru, India*
- Engineered and validated structured datasets from **20,000+** website records using **SQL** (joins, window functions) and **Python (Pandas)**, with deduplication, null handling and feature engineering for phishing risk analysis.
- Conducted EDA with **Matplotlib** and **Seaborn** to find high risk URL and domain patterns, improving phishing detection accuracy by **12%**.
- Designed interactive **Power BI** dashboards with drill through filters and KPI tracking, presenting weekly risk trends to IT security and management.

Projects

E-Commerce Analytics Pipeline with GenAI Insights Assistant | *Python, PostgreSQL, SQL, Tableau, Chroma, Claude API*
- Designed a **PostgreSQL** schema from raw e-commerce data with **SQL** analyses and **Tableau** dashboards.
- Added a **RAG** layer for natural-language querying, embedding metric summaries with **Chroma**.
- Built a QA loop with the **Claude API**, tuning retrieval depth to remove hallucinated answers; published on GitHub.

A Data-Driven Approach for Detecting AI-Generated Images | *Python, PyTorch, ResNet-50, EfficientNet-B3, FastAPI, React*
- Built an AI image forensics system (**ResNet-50, EfficientNet-B3**, CNNs): **91.4% accuracy**, **0.968 AUC**, ˜28K images.
- Designed a wavelet based frequency domain CNN; ablation studies improved performance by **4.86%** over a pixel space baseline.
- Deployed a **React and FastAPI** app with ensemble inference, Grad-CAM explanations and **Claude API** report summaries.

RideAI | *Java, Spring Boot, Next.js, Kafka, Redis, Docker, GitHub Actions, Testcontainers, Sentry*
- Built a real time ride hailing platform with **Java, Spring Boot**, **Next.js** and **Kafka** event streaming.
- **Redis GEO** driver matching sustained ˜**4,900 GPS updates/s** at **p95 34 ms**, **0% errors**, across 100 simulated drivers.
- Set up CI/CD with **GitHub Actions**, integration tests with **Testcontainers**, **Docker** and **Sentry** monitoring.

Morse Code-Based Secure Authentication using Eye Blink Recognition | *Python, OpenCV, Haar Cascade, LBPH*
- Built a computer vision system that translates eye blink patterns into Morse code for secure user verification.
- Implemented face and eye detection with **OpenCV Haar Cascades** and **LBPH** for real time recognition, with blink-pattern logic to reduce false authentication attempts.
- Recognized and funded by the **Karnataka State Council for Science and Technology (KSCST)**.

Technical Skills

Languages & Databases: Python, SQL, Java, JavaScript, R, C, PostgreSQL, MySQL, MongoDB, Apache Spark, Hadoop, Databricks
ML & GenAI: PyTorch, TensorFlow, Scikit-learn, XGBoost, LLMs, RAG, LangChain, Prompt Engineering, Agentic AI
Analytics & Visualization: Pandas, NumPy, Tableau, Power BI, Matplotlib, Seaborn, Plotly, Shiny
Backend & DevOps: FastAPI, Flask, Spring Boot, React, Next.js, Kafka, Redis, Docker, Git/GitHub, GitHub Actions, AWS

Certifications

AWS Certified Cloud Practitioner • Generative AI with LLMs (DeepLearning.AI) • Agentic AI (DeepLearning.AI) • Streamlit GenAI Apps (DeepLearning.AI) • Python for Data Science, AI & Development (IBM) • Python for Everybody (U. Michigan)